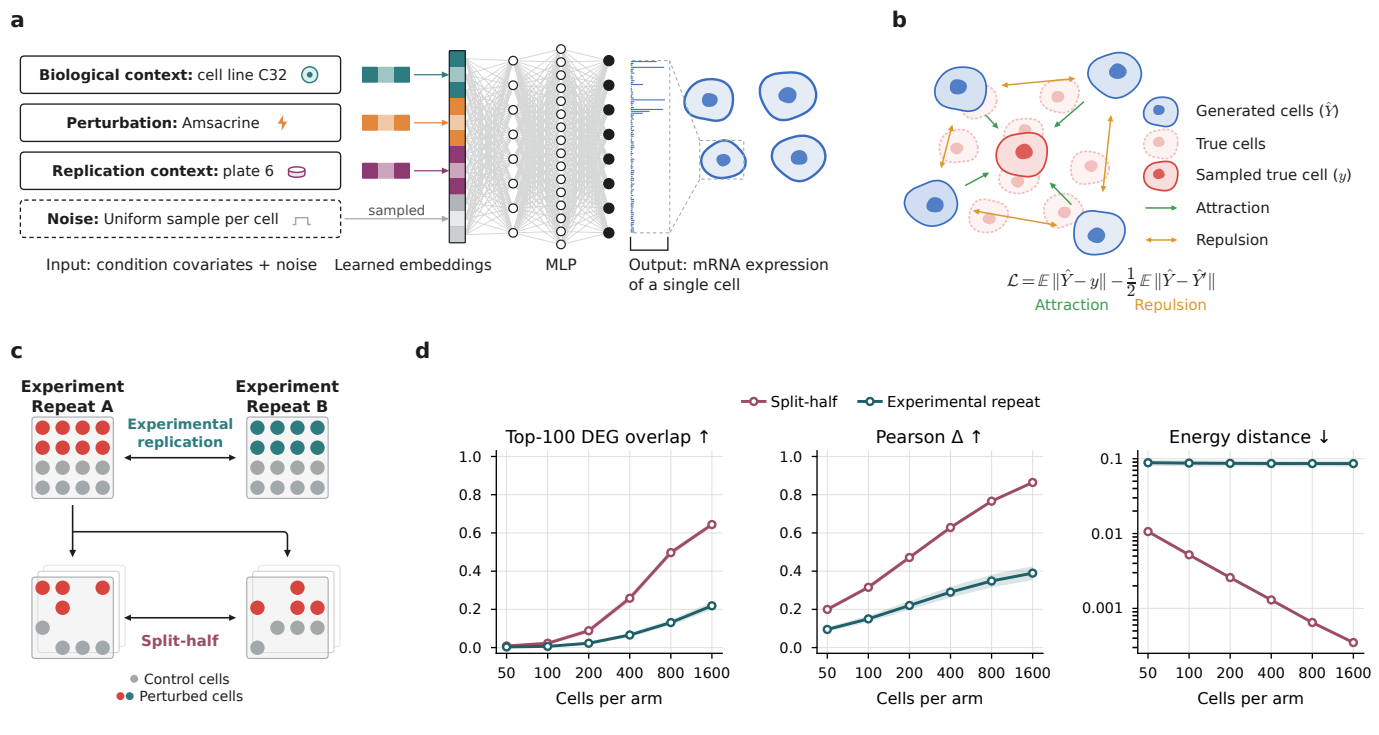

a
Biological context: cell line C32
Perturbation: Amsacrine
Replication context: plate 6
Noise: Uniform sample per cell
sampled
Input: condition covariates + noise
Learned embeddings
MLP
Output: mRNA expression of a single cell
b
Generated cells ($\hat{Y}$)
True cells
Sampled true cell (y)
Attraction
Repulsion
$\mathcal{L} = \mathbb{E}\|\hat{Y} - y\| - \frac{1}{2}\mathbb{E}\|\hat{Y} - \hat{Y}'\|$
Attraction
Repulsion
c
Experiment Repeat A
Experiment Repeat B
Experimental replication
Split-half
Control cells
Perturbed cells
d
Split-half
Experimental repeat
Top-100 DEG overlap ↑
Pearson Δ ↑
Energy distance ↓
Cells per arm
Cells per arm
Cells per arm